INTRODUCTION

Earlier lessons addressed a firm's short-term activities and longer-term capital investment decisions. We now turn to the last part of the balance sheet: long-term debt and equity financing, known as a firm's capital structure. The first lesson introduces the basic objective of most managers when choosing a capital structure: minimizing the firm's weighted-average cost of capital. The second lesson considers the internal and external factors that influence a firm's choice of—and investors' willingness to offer—debt versus equity financing. While capital structure seems like an important decision for boards and managers, it is the present value of future cash flows, rather than a firm's capital structure, that primarily drives a firm's value, a central insight in the influential work of Franco Modigliani and Merton Miller. In the third lesson, we explore the simplifying assumptions used by Modigliani and Miller to demonstrate the irrelevance of capital structure to firm value, and then we relax these assumptions to show the impact of both taxes and the cost of financial distress. In the final lesson, we discuss optimal and target capital structures for issuers.

Category
Capital Structure

Learning Module Overview

- An issuer's cost of capital is composed of its cost of debt and equity, which are defined as its investors' required rates of return on debt and equity financing. An issuer's cost of capital is estimated using a weighted average of the costs of debt and equity, using either current market value or management's target weights of each type of financing as the weights.
- Issuers generally aim to minimize their weighted-average cost of capital and to match the duration of their assets and financing. Managements' target capital structures are usually stated using book values or indirectly through financial leverage ratios, such as a maximum ratio of debt or net debt to EBITDA or a minimum credit rating.
- While management has some influence, the total amount and type of financing needed or the weights in the WACC calculation often depends on the issuer's business model (e.g., capital intensive or capital light) and on the company's life cycle stage.
- The component costs of debt and equity are determined by top-down factors, such as financial market and industry conditions, and by issuer-specific factors, including the stability of revenues and operating and financial leverage.
- Modigliani and Miller (MM) showed, under a restrictive set of assumptions including no taxes, that an issuer's capital structure is irrelevant to firm value. MM relaxed the assumptions by considering corporate taxes, financial distress, and bankruptcy costs and showed that capital structure does matter, although far less than an issuer's future cash flows, for firm value.
- Under MM's static trade-off theory of capital structure, the optimal capital structure occurs where the tax benefit of debt equals the financial distress costs associated with debt.
- The pecking order theory of capital structure is an alternative to the static theory and suggests that a firm will use internal financing as much as possible. If external financing is needed, the firm prefers private debt over public debt and will limit the use of equity financing if possible.

These initial questions are intended to help you gauge your current level of understanding of this learning module.

1. In computing WACC, the cost of equity is higher than the cost of debt because the:

A. cost of debt is set by management.
B. distributions to shareholders are tax deductible.
C. debt investors take less risk than equity investors.

Hide Solution

Solution:

C is correct. Debt is less risky than equity because it has a priority, contractual claim on the firm's cash flow. Additionally, some debt is secured with an underlying asset. In contrast, equity is a residual claim.

A is incorrect because an issuer's cost of debt, like its cost of equity, is determined by financial market participants.

B is incorrect because, generally, distributions to shareholders, such as dividends, are not tax deductible while interest expense is tax deductible in many jurisdictions.

2. Nutry, Inc., has a capital structure of 30% debt and 70% equity, and interest expense is tax deductible. Debt investors require a before-tax return of 5%, and equity investors' required return is 10%. If the marginal corporate tax rate is 20%, the WACC is closest to:

A. 5.9%.
B. 8.2%.
C. 8.5%.

Hide Solution

Solution:

B is correct. Nutry's WACC is calculated as follows:

WACC = (Weighting of debt × Cost of debt) + (Weighting of equity × Cost of equity)

= (0.3)(5%)(1 – 0.2) + (0.70)(10%) = 8.2%.

Thus, the WACC for Nutry is 8.2%. The cost of debt is stated on an after-tax basis because interest expense is tax deductible in Nutry's jurisdiction.

3. The optimal capital structure is determined where the benefit of the debt tax shield is offset by the cost of financial distress under the:

A. pecking order theory.
B. free cash flow hypothesis.
C. static trade-off theory of capital structure.

Hide Solution

Solution:

C is correct. The static trade-off theory of capital structure incorporates both the value-enhancing effect of the tax shield and the value-reducing impact of the costs of financial distress. At the optimal level of debt, the financial distress cost equals the tax benefit of debt.

The pecking order theory states that firms use internally generated funds first because there are no floatation costs or negative signals. If more funds are

The free cash flow hypothesis argues that higher debt levels discipline managers by forcing them to manage the company efficiently and use cash wisely so the company can make its interest and principal payments.

4. True or False: Managers cannot precisely estimate the optimal capital structure, but they often establish a target capital structure. An issuer's actual capital structure may differ from its target based on business and financial market conditions.

A. True
B. False

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Solution:

A is correct. Since management cannot estimate the optimal capital structure in practice, it instead sets a target capital structure. The actual capital structure may deviate from the target for several reasons. First, the firm may be able to issue debt at a favorable rate, so management takes advantage of these opportunities. Second, changing market values of the firm's debt and equity may cause the firm's actual capital structure to differ from its target. Transaction costs make it costly to constantly adjust to the changing market values.

5. The amount and type of financing needed or the weights in the WACC calculation depend on the issuer's:

A. business model.
B. financial leverage.
C. proportion of fixed cost to total costs.

Hide Solution

Solution:

A is correct. The amount and type of financing needed or the weights in the WACC calculation depend on the business model and the stage in the company's life cycle. Some businesses require large amounts of assets and are capital intensive. Other business models require less assets and are capital light.

B and C are incorrect because the proportion of fixed assets to total costs (operating leverage) and financial leverage are issuer-specific factors that influence the component costs of debt and equity.

6. Under Modigliani and Miller (MM), if one assumes no taxes and no financial distress cost, among other assumptions, the value of the company is:

A. determined by its capital structure.
B. determined solely by its expected future cash flows.
C. set so the value of levered company is greater than that of the unlevered company.

Hide Solution

Solution:

B is correct. MM showed that under a set of restrictive assumptions, including zero taxes, the firm's value is unaffected by its financing mix or capital structure. It is the firm's cash flow that is the primary determinant of value. If the market value of the company is not affected by its financing mix, then the value of the levered firm is equal to the value of the unlevered firm.

Discussion

What do you want to discuss... (required)

No messages to display

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